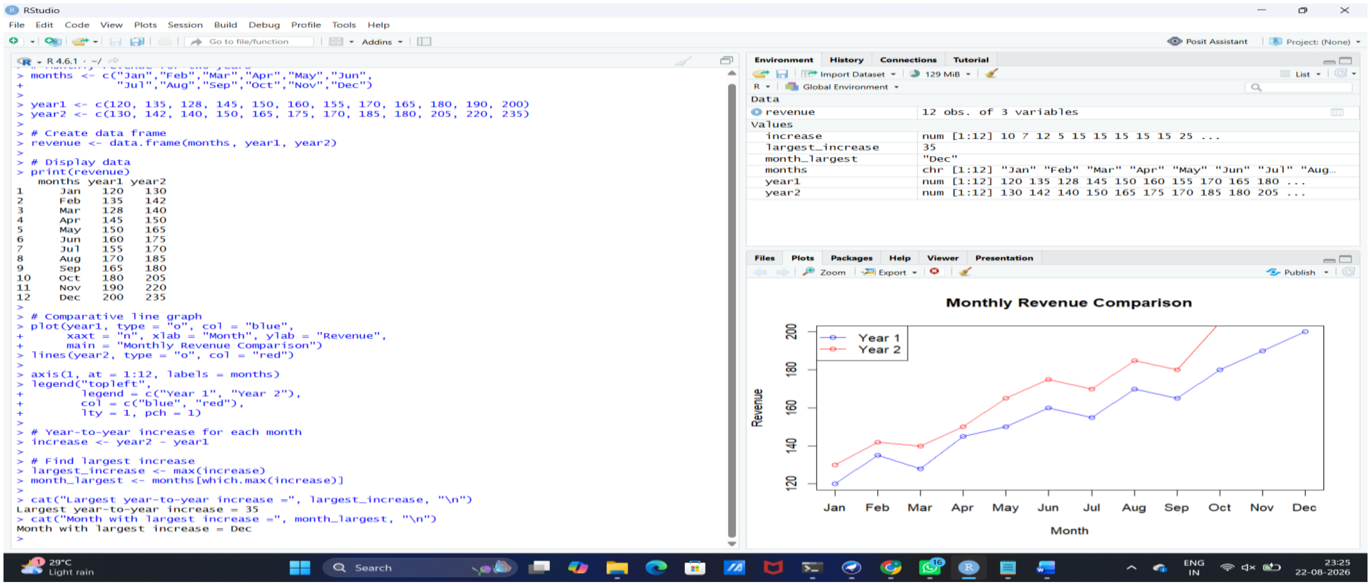
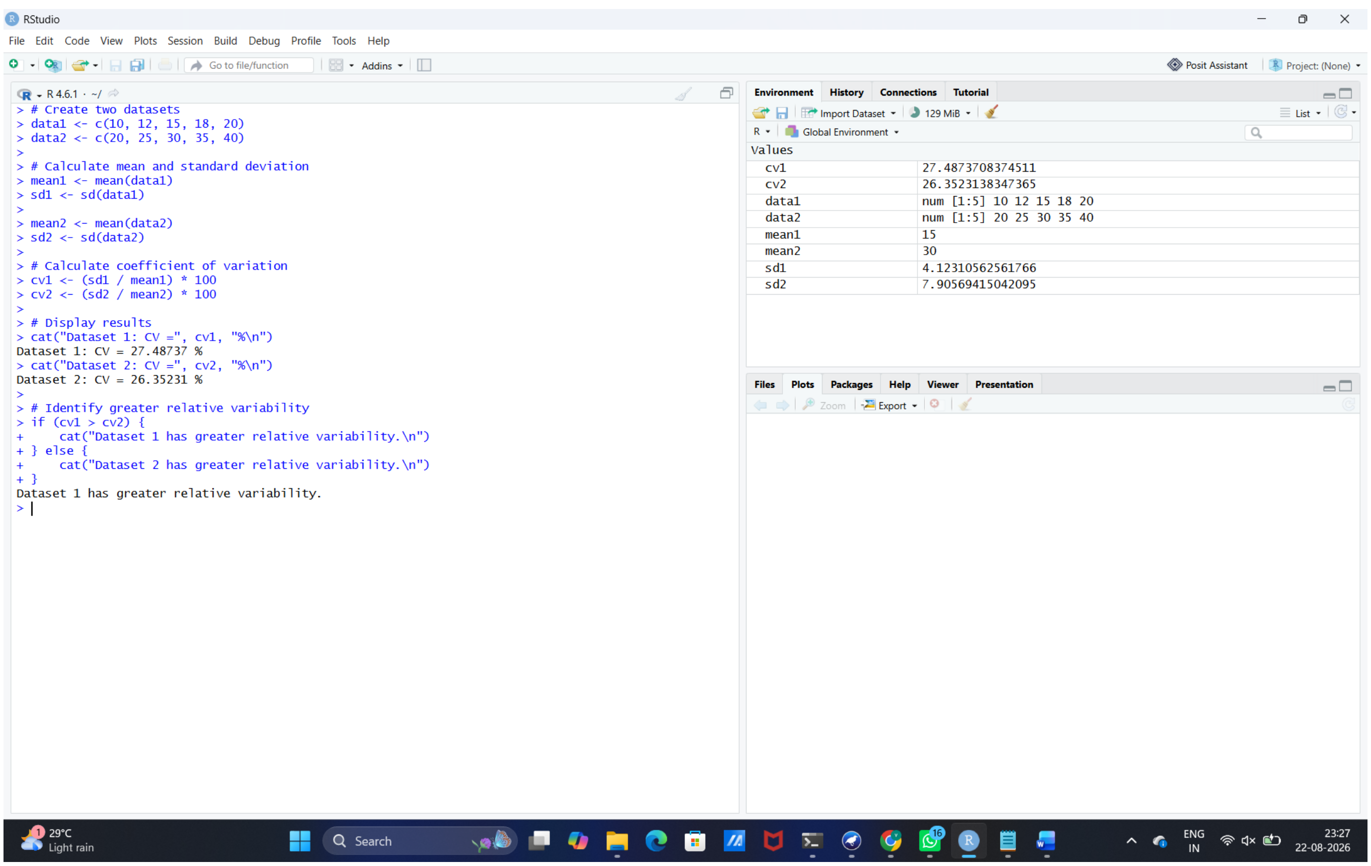
SET 18

1. R: Create monthly revenue for two years; draw a comparative line graph and find the largest year-to-year increase.



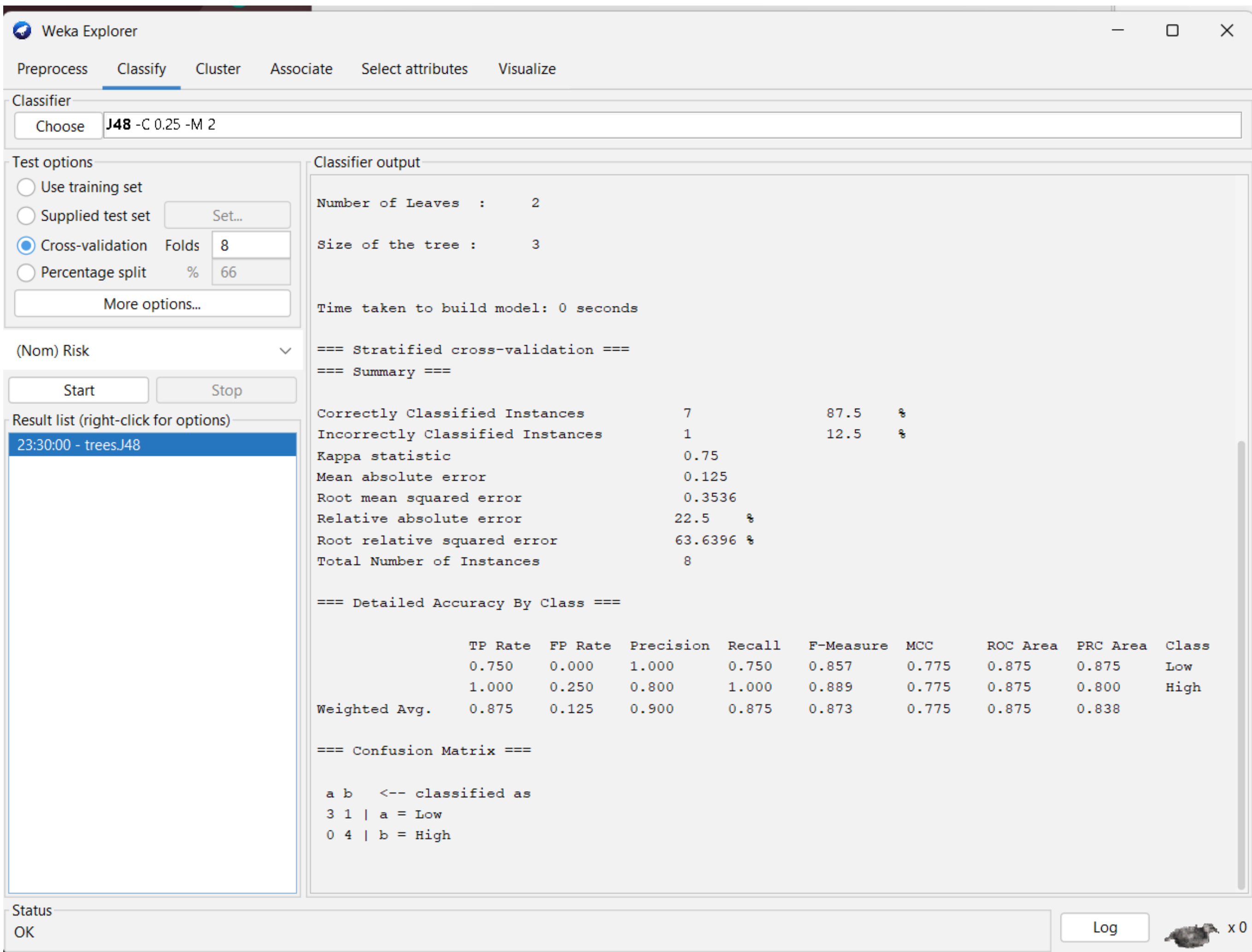
2. R: Calculate coefficient of variation for two datasets and identify greater relative variability.



3. WEKA: Create medical-screening ARFF data (BloodPressure, Cholesterol, Age, Risk); apply J48 and visualize.

Dataset: Medical Screening

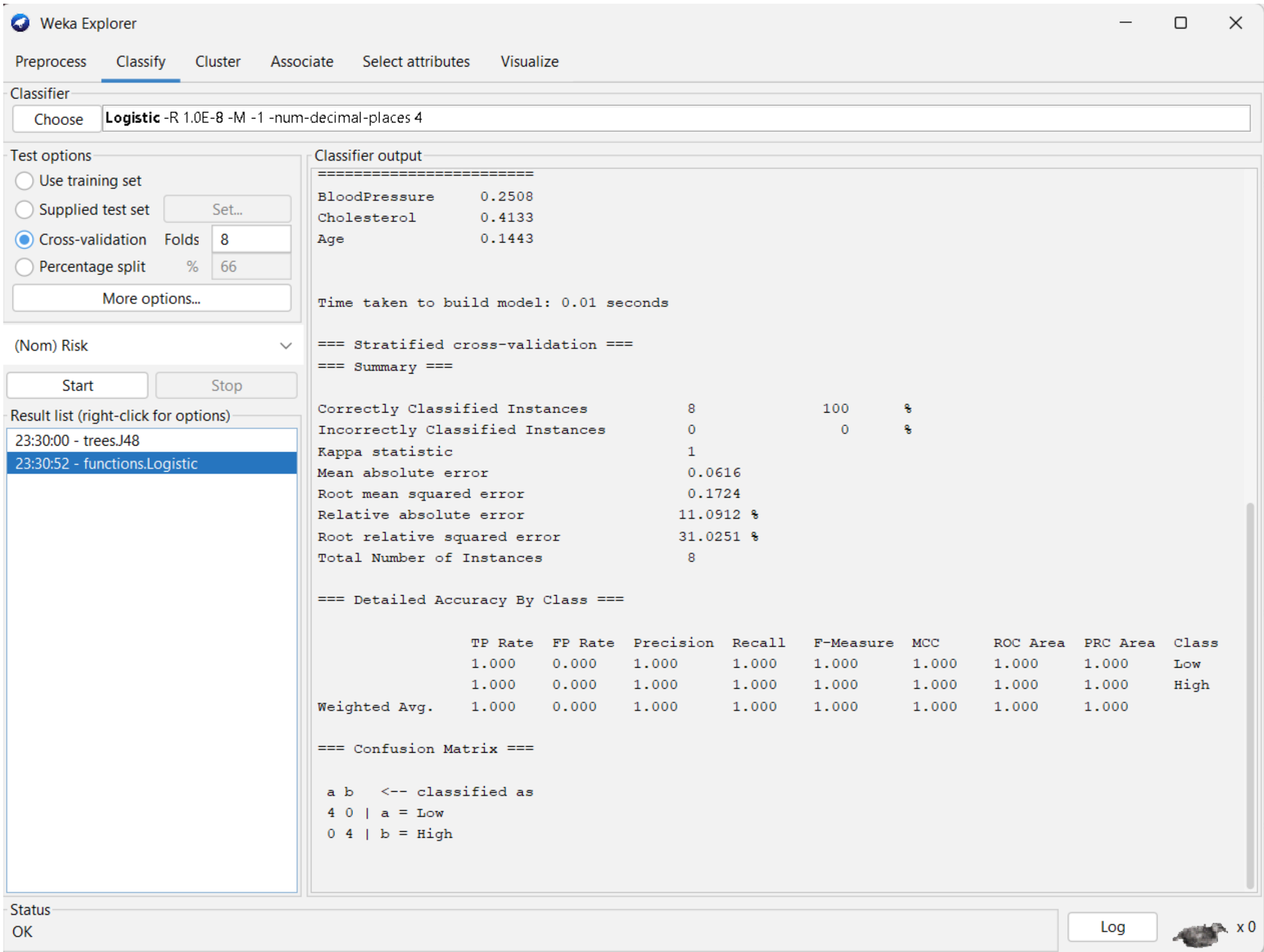
```
@relation medical_screening
@attribute BloodPressure numeric
@attribute Cholesterol numeric
@attribute Age numeric
@attribute Risk {Low,High}
@data
118,170,25,Low
135,220,45,High
125,190,35,Low
145,240,55,High
130,210,42,High
115,165,28,Low
150,250,60,High
122,180,32,Low
```



4. WEKA: Apply Logistic classification to the medical data and compare accuracy and ROC area with J48.

Dataset: Medical Screening – Logistic

Use the same Medical Screening ARFF dataset from Set 18 Q3.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
8
Percentage split
%
66
More options...
(Nom) Risk
Start
Stop
Result list (right-click for options)
23:30:00 - trees.J48
23:30:52 - functions.Logistic
Classifier output
=======================
BloodPressure 0.2508
Cholesterol 0.4133
Age 0.1443
Time taken to build model: 0.01 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 8 100 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0.0616
Root mean squared error 0.1724
Relative absolute error 11.0912 %
Root relative squared error 31.0251 %
Total Number of Instances 8
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Low
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 High
Weighted Avg. 1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000
=== Confusion Matrix ===
a b <-- classified as
4 0 | a = Low
0 4 | b = High
Status
OK
Log
x 0